PHASE 1: IDEATION

EV Market Performance & Insights

Date	20 – October - 2026
Nasscom Student Id	SWUID2026399918
Project Name	Electric vehicle Market Performance & Insights

NASSCOM Project Deliverables: Problem Statement • Empathy Map • Brainstorm & Idea Prioritization

Purpose: This document organizes the ideation deliverables for a Tableau-based analysis of the available electric vehicle dataset. Replace any illustrative assumptions with verified project details where necessary.

1. Problem Statement

1.1 Background

Electric vehicle (EV) datasets can contain several dimensions of market information, including vehicle category, model year, manufacturer, battery technology, driving range, customer ratings, and manufacturing country. Reviewing these fields separately can make it difficult for a viewer to build a consolidated understanding of market patterns.

1.2 Problem

Users who need to explore EV market performance may find it time-consuming to compare multiple categories and performance measures when the information is presented as separate data fields or individual charts. A clear, interactive view is needed to support exploration and comparison of the available dataset.

1.3 Target Users

- Data analysts exploring EV market patterns.
- Business or market teams comparing manufacturers and vehicle categories.
- Researchers or students reviewing EV-related trends in the available dataset.

1.4 Proposed Problem Statement

How might we present the available electric vehicle data in a clear and interactive Tableau dashboard so that users can compare EV categories, annual sales, manufacturers, battery technologies, driving range, customer ratings, and manufacturing countries more efficiently?

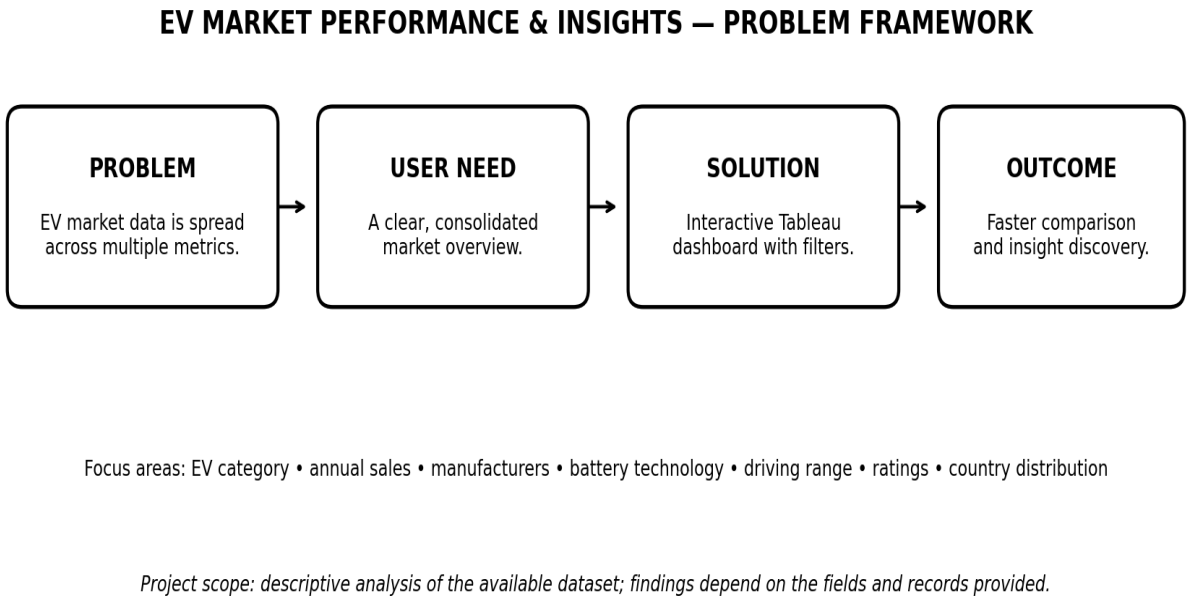
1.5 Project Objectives

- Present the main EV-related measures and dimensions in a single dashboard.
- Compare annual sales trends and manufacturer representation in the dataset.
- Explore the distribution of EV categories and battery technologies.
- Compare average driving range and customer ratings across EV categories.

- Provide filters that let users explore selected categories, manufacturers, or model years.
- Communicate observations supported by the available dataset without making unsupported predictions.

1.6 Expected Outcome

An interactive Tableau dashboard and Story that organize the available EV data into seven visualizations, enabling users to explore patterns and compare selected market dimensions. The dashboard supports descriptive analysis; conclusions are limited to the dataset and fields available.



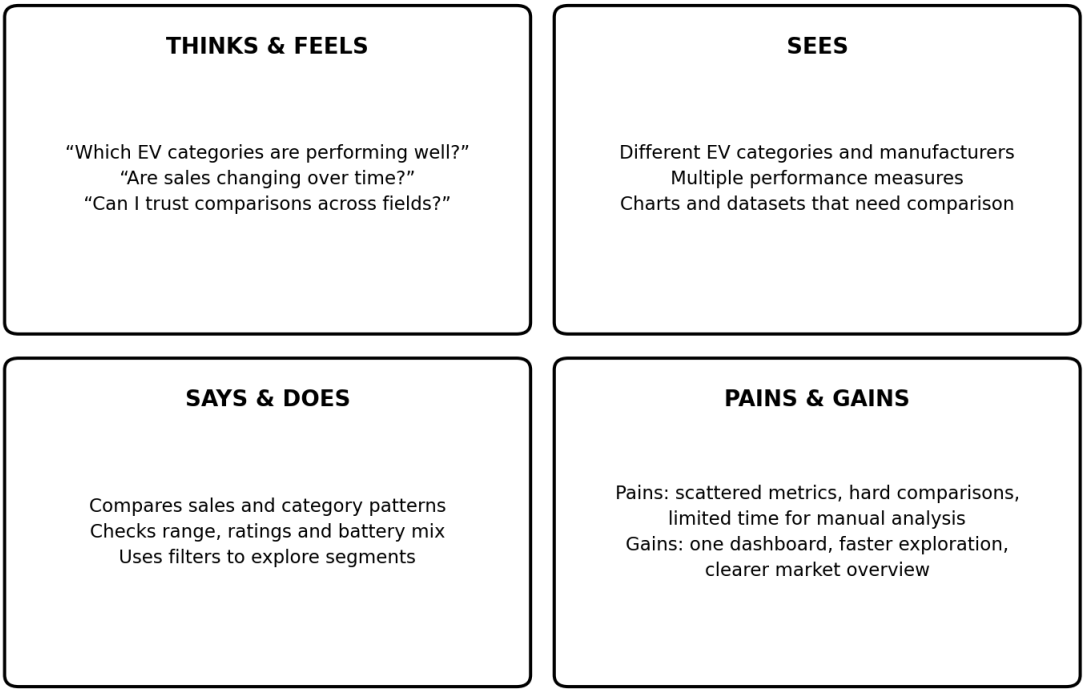
2. Empathy Map

Persona: EV Market Analyst / Business User (illustrative persona created for project planning). This empathy map is a working hypothesis based on the dashboard's intended use, not a result of interviews or a formal user study.

Empathy Area	Illustrative User Perspective
THINKS & FEELS	"Which EV categories are most represented?" "How have annual sales changed?" "Can I compare manufacturers and vehicle performance quickly?"
SEES	EV-related fields covering categories, model years, manufacturers, battery technology, range, ratings, and country. Multiple measures may need to be compared.
HEARS	Requests from colleagues or project reviewers for clear summaries, comparisons, and evidence-backed findings about EV market patterns.
SAYS & DOES	Compares charts, applies filters, looks for differences across categories, and checks whether an observation is supported by the data.
PAINS	Comparisons can be time-consuming when measures are viewed separately; crowded labels or unclear

	chart titles can make insights harder to interpret.
GAINS / NEEDS	A consolidated dashboard, readable visualizations, useful filters, consistent measures, and concise findings that can be explained to others.

EMPATHY MAP — EV MARKET ANALYST (ILLUSTRATIVE PERSONA)



Note: This map is a project-planning hypothesis, not a record of real interviews.

Key Design Implications

- Use clear titles and readable axis/category labels.
- Keep related visualizations together and avoid unnecessary decoration.
- Provide relevant filters and ensure they behave as intended.
- Write findings using actual values or patterns visible in the dashboard.

3. Brainstorm & Idea Prioritization

The team considered several ways to communicate EV market information. Ideas were compared against project relevance, feasibility within the available tools and time, and usefulness to the intended audience.

3.1 Brainstormed Ideas

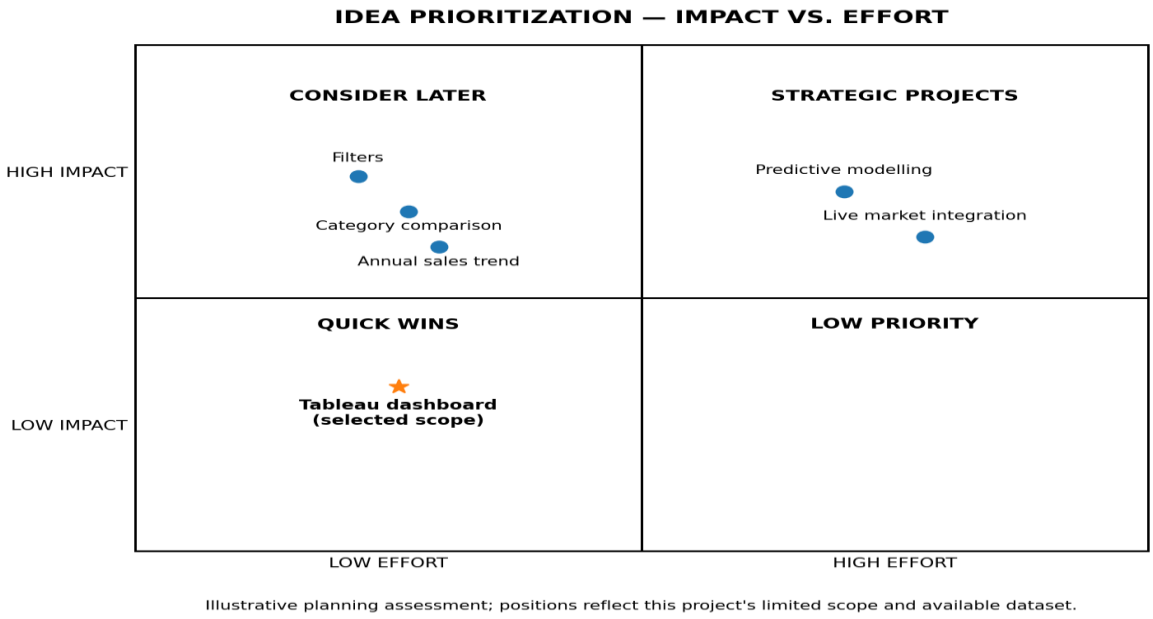
Idea	Potential Benefit	Feasibility / Limitation
Interactive Tableau dashboard	Combines key EV views and enables filter-based exploration.	High; aligns with the project's Tableau-only scope.
Static report with charts	Easy to share and read as a fixed summary.	High, but users cannot interact with the charts.
Predictive EV sales model	Could estimate future sales if suitable historical data and validation were available.	Lower for current scope; requires modelling, validation, and additional assumptions.
Live EV market monitoring	Could show updated information when new data arrives.	Lower for current scope; requires a reliable live data source and refresh setup.
Tableau Story with multiple points	Presents related views in a logical narrative.	High; can be built using the current Tableau workbook.

3.2 Prioritization Criteria

- Relevance: Does the idea answer the project's EV market questions?
- Feasibility: Can it be implemented with the available dataset and Tableau?
- Usability: Can users understand and explore the result easily?
- Scope fit: Can it be completed and documented within the project timeline?

3.3 Selected Idea

The selected solution is an interactive Tableau dashboard supported by a Tableau Story. It is aligned with the project's available dataset and current scope, and it supports descriptive comparisons across the seven visualizations without requiring a separate predictive model or live-data integration.



3.4 Final Rationale

- Uses the existing EV dataset and Tableau workbook.
- Covers the main project questions through seven visualizations.
- Allows interactive exploration using available filters.
- Can be demonstrated and documented without claiming predictive or live-market capabilities.